

Customer Retention & Churn Analysis

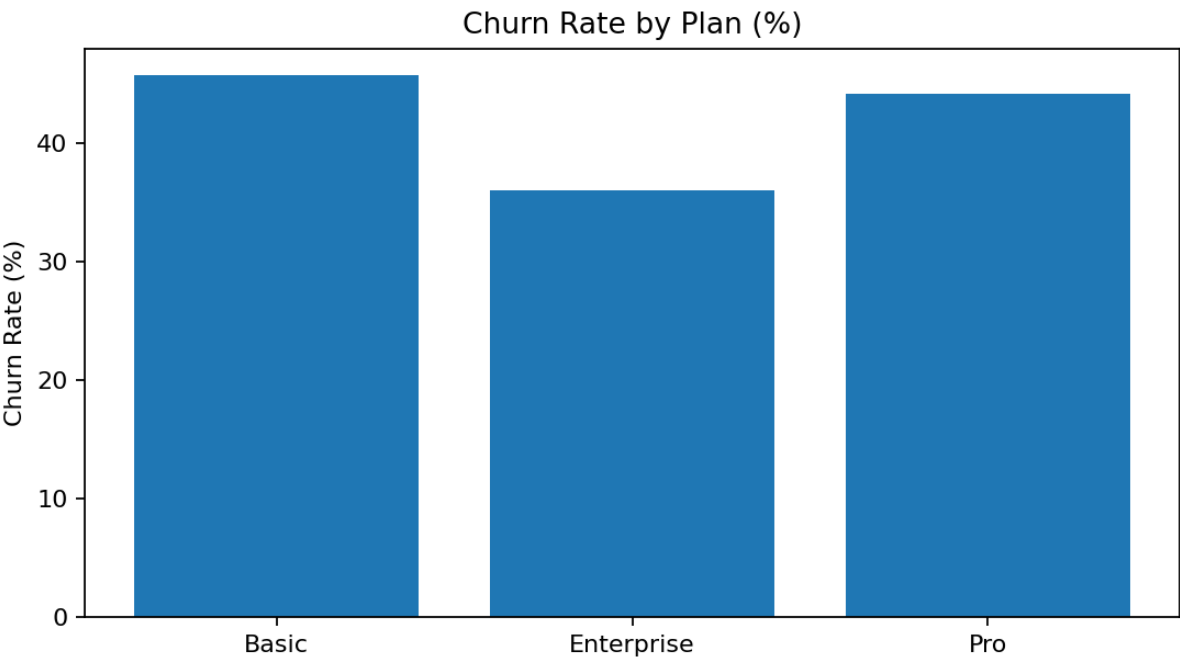
Future Interns - Data Science & Analytics Task 2 (2026)

Executive Summary

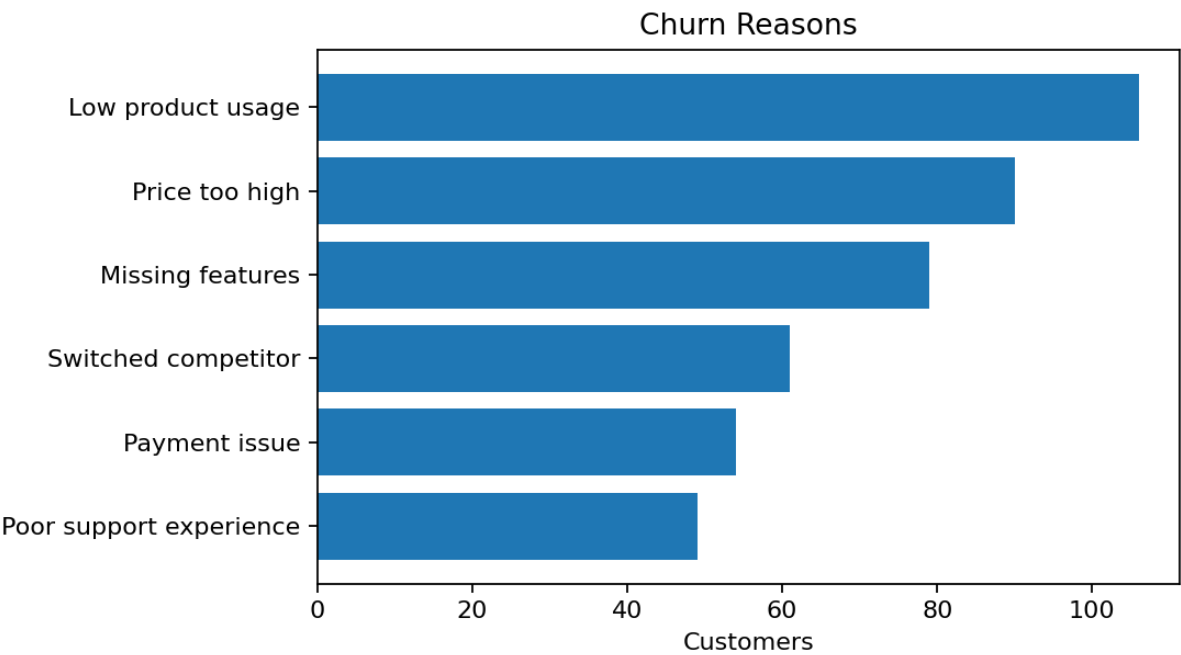
This report analyzes a simulated SaaS/subscription customer dataset with **1,000** customers. The overall churn rate is **43.9%**, with a retention rate of **56.1%**. Average tenure is **4.8 months** and average customer lifetime value is **\$232.32**. The top churn reason is **Low product usage**, indicating that customer engagement and activation should be the main retention focus.

Metric	Value
Total Customers	1,000
Churned Customers	439
Active Customers	561
Churn Rate	43.9%
Retention Rate	56.1%
Average Tenure	4.8 months
Average CLV	\$232.32

Key Visual Insights

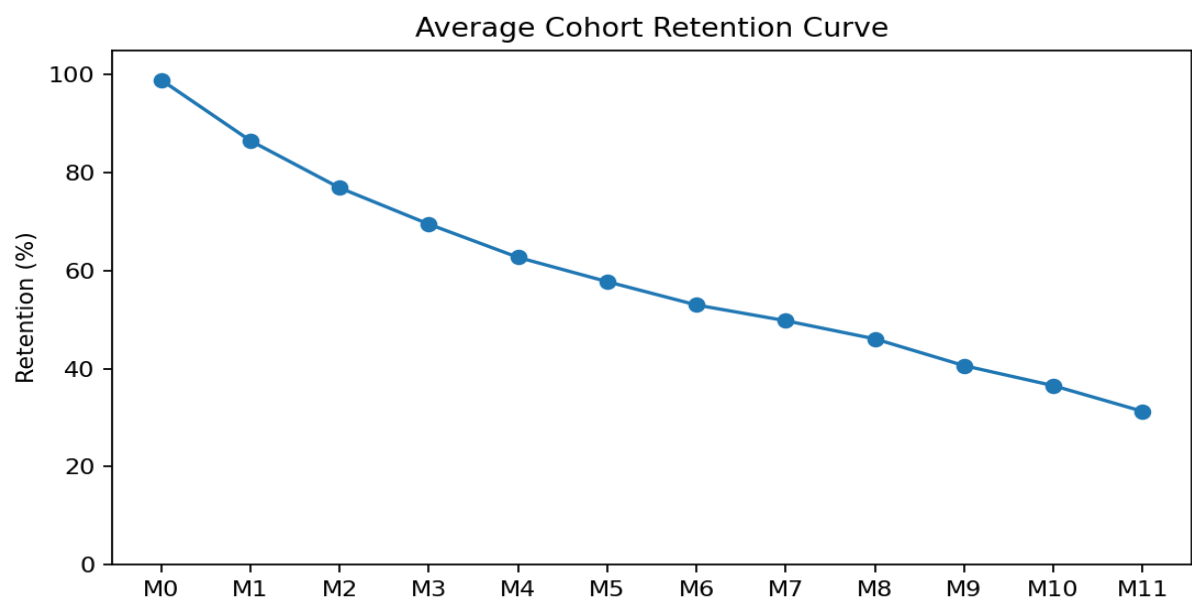
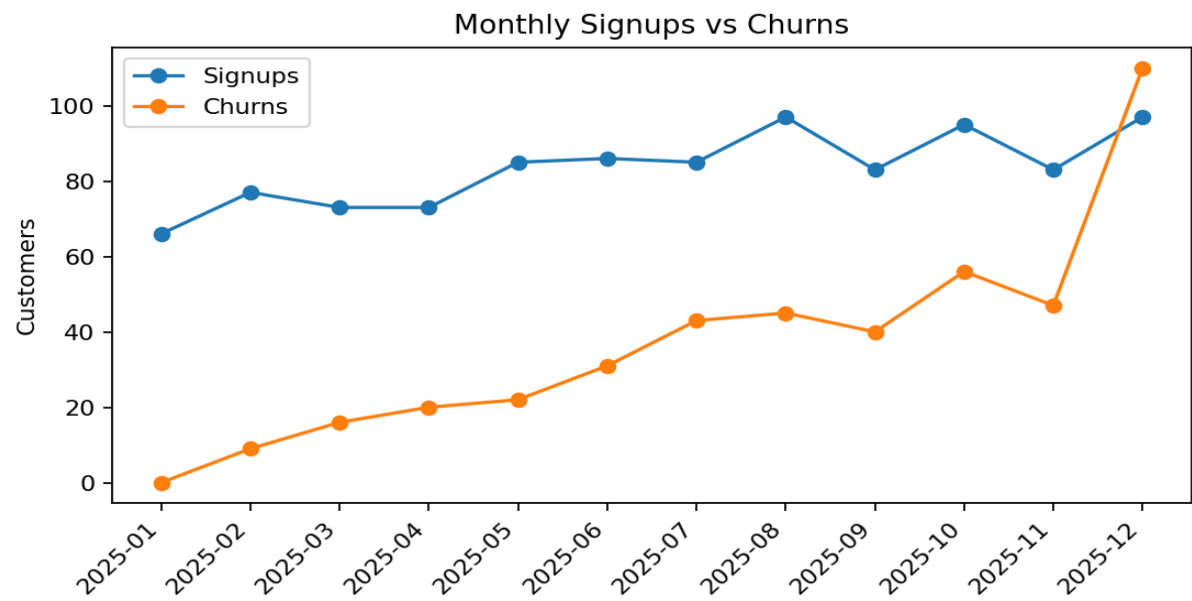


Churn Reasons



The largest churn category is low product usage, followed by price sensitivity and missing features. This points to a combined need for stronger activation, better value communication, and product feedback loops.

Retention Trend and Cohort Analysis



Cohort retention declines most during early customer life, so onboarding, usage activation, and first-value milestones are the highest priority levers.

Segment Findings

Segment	Customers	Churned	Churn Rate	Avg CLV
Basic	518	237	45.8%	\$88
Enterprise	136	49	36.0%	\$769
Pro	346	153	44.2%	\$238

Recommendations

- Reduce low-usage churn:** Launch a 14-day onboarding journey, trigger usage-based nudges, and alert customer success when usage score falls below 50.
- Protect the Basic plan segment:** Basic customers are the largest segment and have the highest churn rate, so tailor setup guidance and value education to this group.
- Improve acquisition quality:** Referral and partner channels retain better than paid/social channels; adjust budget toward higher-retention sources.
- Create support-risk intervention:** Flag customers with 3+ support tickets for proactive outreach and fix recurring product/support issues.
- Reduce payment-related churn:** Use card-expiry reminders, billing retry logic, and proactive payment support to recover avoidable churn.

Project Note

The dataset is simulated and anonymized for portfolio use. The accompanying Excel workbook contains the full customer table, formulas, cohort analysis, segment analysis, dashboard, and recommendations.